SPACE HUNTER LAZY WORKFLOW

Version: Part 1 – RELEASE v1.2 / Part 2 – RELEASE v1.2

LICENSE:

Both scripts are released under the MIT License. Copyright © 2026 Georg G Albrecht. You are free to use, modify, and redistribute these scripts with the **exception** of AutoColor which requires to be kept in its original code state and requires a separate written agreement by the author. See the included LICENSE.PDF files downloaded with the .Zip file for full terms.

USAGE:

The scripts run on Windows only and are primarily used for processing **OSC data** collected with broadband, light-pollution or single Dual-Narrowband (DNB) filters without splitting RGB channels into individual grey scale images. If you are using two DNB filters collecting Ha/Oiii and Sii/Oiii(Hb) to combine channels into Hubble palette images, this is not the right script for you! However, I might at some point in the future come up with other scripts...

FUNCTIONALITY:

These scripts were designed to automate the repetitive, non-artistic steps of image processing using the latest available processes in PixInsight. Rather than manually calling scripts and processes one by one, the workflow handles everything automatically in two parts — freeing you to focus on the decisions that actually require your judgment.

Part 1 takes a pre-cropped image from plate solving through to background extraction, running each step in sequence without manual intervention. Once you've completed your background extraction and are ready to continue. **Part 2** picks up from there and carries the image through to a stretch-ready, star-separated result.

Both parts offer flexibility — steps can be enabled or disabled, and users who prefer older background extraction methods (such as GC or DBE) over MGC can easily switch. Equally, AutoColor and StarNet2 have been added to part 2 as alternatives for SPCC and StarXTerminator.

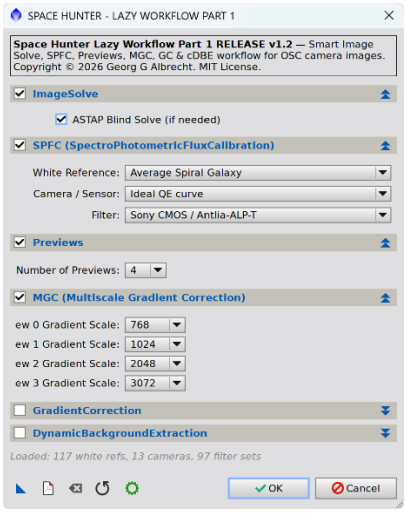
REQUIREMENTS:

- Windows operating system
- PixInsight 1.8.9.3 or later
- Mars Databases (can be on C:\ or other drive *)
- ASTAP and D50 star data installed (preferable under C:\Program Files\astap *)
- BlurXTerminator (BXT), StarXTerminator (SXT), NoiseXTerminator (NXT) **, StarNet2 and AutoColor
- Start in PixInsight from a pre-cropped image ***

* If file location can not be found automatically the first time the scripts start, it will ask you to find the location manually. Once you have designated the file location, the scripts will save path for future runs.

** If BXT and NXT are not installed, the script passes to the next step. If SXT is not installed, the script looks for StarNet2.

*** If you stack subs in WBPP with Auto Crop and Astrometric Solution enabled, start the script without pre-cropping. ImageSolve is automatically skipped.

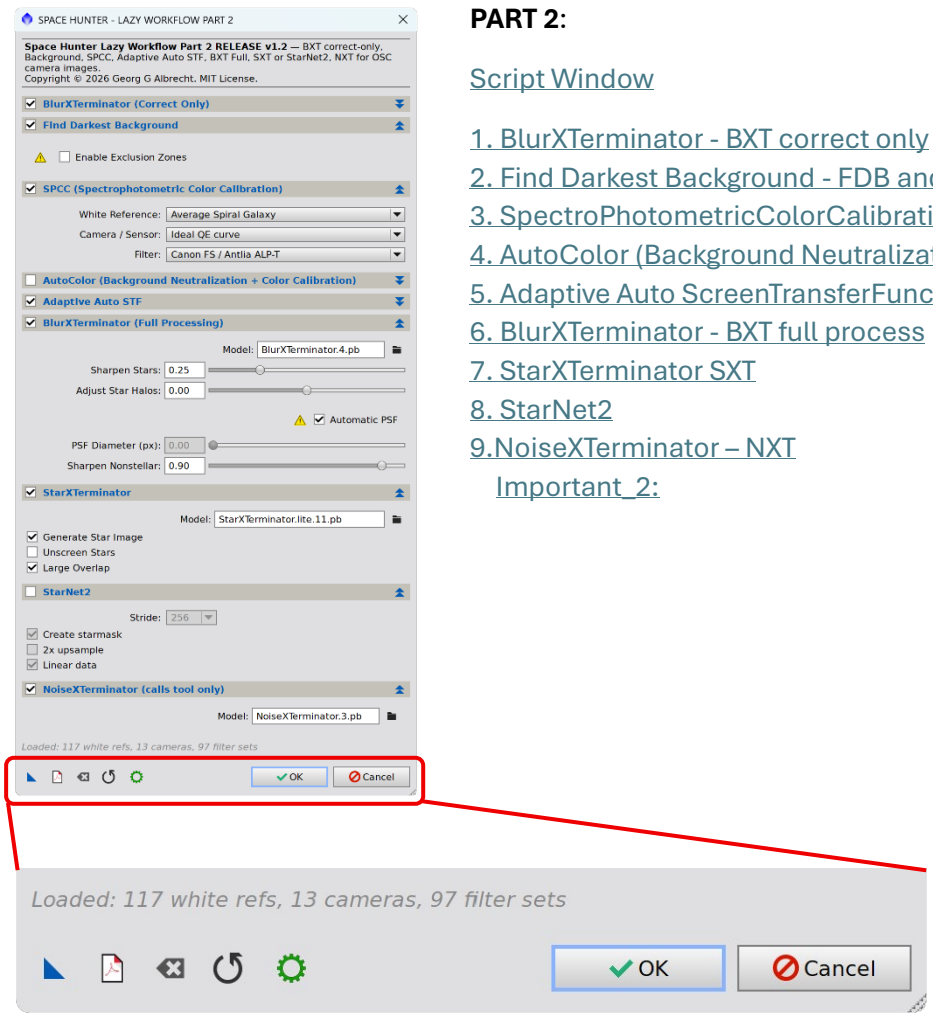


PART 1:

Script window

1. Image Solve with ASTAP blind solve
2. SpectroPhotometricFluxCalibrartion - SPFC
3. Previews
4. MultiscaleGradientCorrection - MGC
5. GradientCorrection - GC
6. Custom DynamicBackgroundExtraction - cDBE

Important_1:



PART 2:

Script Window

1. BlurXTerminator - BXT correct only
2. Find Darkest Background - FDB and Exclusion Zones
3. SpectroPhotometricColorCalibration – SPCC
4. AutoColor (Background Neutralization + Color Calibration
5. Adaptive Auto ScreenTransferFunction - AASFT
6. BlurXTerminator - BXT full process
7. StarXTerminator SXT
8. StarNet2
9.NoiseXTerminator – NXT

Important_2:

From left to right: New instance > User guide > Deselect all WorkFlow steps > Reset WorkFlow steps to default > Reset Gear Selection to default

Part 1: Plate Solve → SPFC → Previews → MGC → GC → cDBE

Workflow & Equipment Selection Dialog. Collapsible sections let the user enable/disable workflow steps. Pre-enabled steps: Plate Solve, SPFC, Previews, MGC.
Disabled steps: GC and cDBE. In Image Solve, ASTAP is pre-enabled but can be disabled if user knows image has required basic solving information. In SPFC, white reference, camera sensor and filter databases (XSPD files) are selected. In Previews, user determines how many previews are needed (up to 4). In MGC each preview(s) gradient scale value for the background model is selected. GC has two preset models and cDBE has basic preset values. Both are by default disabled.
Dependencies/Independencies: If all steps are disabled, the following steps can be enabled and executed on their own: Image Solve with or without ASTAP, Previews. SPFC can only be executed with Image Solve active. MGC can only be executed with Image Solve and SCFC active. If GC is selected, SPFC and MGC are automatically disabled. Image Solve is enabled. If cDBE is selected, SPFC, Previews and MGC are automatically deselected. ImageSolve is enabled.

Step 1 — Smart Plate Solve. (default - Enabled) Checks if the image already has an astrometric solution. If yes, it skips ahead. If not, it examines FITS keywords to determine what's missing. If coordinates and image scale exist, it goes straight to ImageSolve's window. If they're missing, it runs ASTAP (default Enabled) first to inject WCS keywords. If the image is missing multiple components, ASTAP blind solver might kick in (10-20 sec). Once finished, ImageSolve window will open and **you are required to click OK to preceed**. The astrometric solution created here is required by SPFC and in Part 2 by SPCC (both need plate-solved coordinates to match catalog stars).

Step 2 — SPFC (SpectroPhotometricFluxCalibration). (default – Enabled) Using the equipment selections from the Script's window, it applies spectrophotometric flux calibration with the chosen camera QE curve, filter transmission curves, and white reference. This linearizes the color response of the image

Step 3 — Create Previews. (default – Enabled) Creates 1–4 full-size preview copies of the image. These serve as test canvases for MGC so the user can compare different gradient correction strengths without modifying the original. The default is to create 4 previews. If script runs default, the previews will be called BM_1, BM_2... for Background Model and show the individual applied value. If GC is selected, the previews are named GC, GC1... If cDBE runs, no previews are needed or created and if Previews are executed alone, the created previews will be called Preview0, Preview1... [2]

Step 4 — MGC (MultiscaleGradientCorrection). (default – Enabled) Runs MGC on each preview with a different user-selected gradient scale (e.g., 768, 1024, 2048, 3072), using the MARS database for reference. After processing, it opens the MGC tool so the user can inspect results and choose their preferred scale. The user then applies their chosen scale to the main image before moving to Part 2.

Step 5 – GC (GradientCorrection). (default – Disabled) Runs GradientCorrection using Preview0 and Preview1 to produce two background corrected images at preset values. It also creates 2 separate images named GC_gradient_model and GC_gradient_model1 with automatically applied AASTF for the user to determine next steps to finalize background extraction. This process is by default disabled but if MGC fails due to missing MARS databases (southern hemisphere), it will automatically start. The process can also be called on its own. If selected on its own, ImageSolve will be included.

Step 6 – cDBE (custom DynamicBackgroundExtraction). (default – Disabled) This process is added only as an alternative for users preferring to do a manual background extraction. If selected, also here ImageSolve will be activated, just to make sure the image has a valid astrometric solution for further processing. The DynamicBackgroundExtraction tool will open with some pre-selected values and add evaluation boxes to the image. The user can adjust the size of the boxes in the tool and add/move/delete boxes as they see fitting.

! At the end of Part 1, after you ran MGC or GC, you will be left with unneeded open previews. I created a bonus script called KillPreviews. It will delete all previews in the active image.

Some important notes to Part_1: If the script ran through its default settings with ImageSolve applied and you chose to roll back image history either by using reverse arrow or History Explorer, astrometric solution is lost if set back to beginning. The script will start from the beginning if applied again. If image is set back to after ImageSolve ran, ImageSolve will still be activated but you may cancel it and SPFC, GC or cDBE will run thereafter.

Part 2: BXT correct → FDB → SPCC or AutoColor → AASTF → BXT full → SXT or StarNet2 → NXT

Workflow & Equipment Selection Dialog. Collapsible sections let the user enable/disable workflow steps. Pre-enabled steps: BXT Correct Only, Find Darkest, SPCC, Adaptive Auto STF, BXT Full, SXT, and Open NXT. Collapsed by default: BXT Correct Only, Adaptive Auto STF, and StarNet2 (with user parameters). Disabled by default: ⚠ Exclusion Zones, AutoColor and StarNet2.

For BXT, SXT and NXT, you will find box where you can link your preferred AI model. This setting will be saved permanently until you chose another model.

In SPCC, white reference, camera sensor and filter databases (XSPD files) are selected. AASTF runs at preset values. Automatic PSF is enabled by default in BXT Full. SXT and StarNet2 are mutually exclusive — enabling one disables the other. At the end of the workflow the original image is brought back to front, ready for NXT to be applied.

All Gear and WorkFlow settings used for a run, will be saved in the script and reopened the next time the script is called. You can reset both, the WorkFlow or the Gear settings to their default by clicking the corresponding icon at the bottom of script.

Dependencies/Independencies: All steps can be enabled or disabled independently with the following exceptions. SPCC automatically enables Find Darkest Background and Adaptive Auto STF when turned on — if the user wants to run either of those steps on their own, SPCC must be disabled first. BXT Correct Only, BXT Full, and NXT are fully independent and can be run on their own. SXT and StarNet2 are mutually exclusive — enabling one automatically disables the other. ⚠ PSF Diameter in BXT Full is only active when Automatic PSF is unchecked and requires user to obtain necessary value before starting the script (DynamicPSF or PSFImage).

Step 1 — BXT (BlurXTerminator) correct Only. (default – Enabled) Runs BXT in correction-only mode — no sharpening, just optical aberration correction. This cleans up star shapes before color calibration. BXT is automatically skipped if not installed.

Step 2 — Find Darkest Background. (default – Enabled) Uses a gradient descent algorithm with 50 random starting paths to find the darkest, most uniform background region in the image. Creates a "Background" preview that SPCC uses as its background reference ROI. Supports optional exclusion zones (previews named "Exclude*") to avoid dark nebulae being mistakenly identified as background.

⚠️ **Exclusion Zones:** (default – Disabled) Before launching script Part 2, if any exclusion zones are needed, the user must manually draw preview rectangles over any dark structures (such as dark nebulae) they want excluded from the background search and rename each preview to start with "Exclude" (e.g., Exclude1, Exclude2) **I created a bonus script called NamePreviews that takes care of the renaming. Run beforehand the KillPreviews, if any previews from Part 1 were left.** The exclusion previews themselves are deleted after the search completes, leaving only the resulting Background preview.

Step 3 — SPCC (SpectroPhotometricColorCalibration). (default – Enabled) Uses the same equipment selections as Part 1's SPFC (white reference, camera, filter) together with the Background preview as its ROI. Matches catalog star spectra to calibrate the image's color balance. It requires the plate-solved astrometric solution from Part 1 or ImageSolve needs to be manually applied.

Step 4 — AutoColor (BackgroundNeutralization + ColorClaibration) - Copyright (c) 2016 Hartmut V. Bornemann. (default – Disabled) This script was added as an alternative to SPCC. It does exactly what is described above in an elegant and simple way. If manually selected, Find Darkest Background and SPCC are automatically disabled.

Step 5 — AASTF (Adaptive Auto ScreenTransferFunction) [1]. (default – Enabled) Applies an adaptive automatic screen stretch, similar to the STF function to the linear image so the user can visually assess the image after color calibration. It works on both, unlinked and linked linear images. This is a display stretch only — it doesn't modify pixel data.

Step 6 — BXT (BlurXTerminator) full. (default – Enabled) Runs BXT with the user's chosen sharpening settings (star sharpening, halo adjustment, nonstellar sharpening, PSF diameter). This is the main deconvolution/sharpening pass. ⚠️ If Automatic PSF is disabled, you either need to guess a PSF diameter value or gadder proper information through DynamicPSF or PSFImage before running the script. BXT is automatically skipped if not installed.

Step 7 — SXT (StarXTerminator). (default - Enabled) Removes stars from the image. Configurable options include generate star image, unscreen mode and tile overlap size. If generate star image is selected, it creates a star only image named “stars” and brings the starless image back to front. SXT is automatically skipped if not installed and script looks for StarNet2 instead.

Step 8 — StarNet2 (default – Disabled) Configurable options include stride, create starmask, 2x upsample, and linear data. If create starmask is selected, it creates a star only image named “stars” and brings the starless image back to front. StarNet2 must be installed separately via the PixInsight repository at https://pixinsight.starnetastro.com/. If manually selects, SXT will be disabled.

Step 9 — NXT (NoiseXTerminator). (default – Enabled) Opens NXT with default settings at the end of the workflow. With the original starless image active, the user can drag the instance triangle from the NXT dialog onto the image to apply noise reduction. NXT is automatically skipped if not installed.

Some important notes to Part_2: Although some safeguards have been added to the script, reversing an image further back than SPCC will erase saved process memory and applying the scrip a second time without disabling already processed workflow steps will alter/damage image data.

At this point, your image should be ready to be stretched and your creative work shall begin.

(1) The Adaptive Auto STF used in these scripts is based on PixInsight's standard Screen Transfer Function methodology by Juan Conejero / Pleiades Astrophoto, with adaptive enhancements for brightness-dependent targeting and cross-channel noise blending.

$$MTF(x,m) = \begin{cases} 0, & x = 0 \\ 0.5, & x = m \\ 1, & x = 1 \\ \frac{(m-1)x}{(2m-1)x-m}, & \text{otherwise} \end{cases}$$

(2) I created an additional scrip called (+Previews) which allows you to create additional previews either for Back ground Models, GradientCorrection or just plain Previews in full size. If you need more Background Model previews to refine your extraction model, you can add them with this scrip easily after Part 1 ran. The same applies for GC processing. You can also just create full-view Previews and...custom named previews, in case there ever would be a reason for. The script is self explanatory.

Thank You Note:

A huge shout-out and thank you to the PixInsight team for their relentless work on this incredible piece of software. This product is based on software from the PixInsight project, developed by Pleiades Astrophoto and its contributors (https://pixinsight.com/). Many thanks.

A special thanks to Franklin Marek at SetiAstro – [Blind Solver 2000] plus team Franklin Marek/Gerrit Erdt – [Find Background] whose scripts inspired me in how to approach and create my ASTAP solver and the FindDarkestBackground process. Also, a big thanks to Nikita Misiura for StarNet2, and to Russell Croman for his invaluable tools BXT, SXT and NXT. Many thanks to Hartmut Bornemann for allowing me to use his AutoColor script. Further, a big thanks to my Beta testers, who pre-tested and returned helpful information and insights. Special thanks to Jeff, Mike and Stefan. Much appreciated.

And to all of you users who downloaded the scripts and use them in your processing, I hope they offer you some value. Thank you and ***Clear Skies***.